

Affine schemes

Preliminaries.

A ringed space is a pair $(X, \mathcal{O}_X)$ such that $\mathcal{O}_X$ is a sheaf of (unital, commutative) rings on X such that the restriction map $V \subseteq U$ of open sets corresponds to (unital) homomorphisms of rings $\mathcal{O}_X(U) \rightarrow \mathcal{O}_X(V)$.

A morphism of ringed spaces between $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ is a continuous map $\varphi : X \rightarrow Y$ and the induced morphism on the sheaf of rings $f^\# : \mathcal{O}_Y \rightarrow \varphi_* \mathcal{O}_X$; where $\varphi_* \mathcal{O}_X$ denotes the direct image / pushforward sheaf given by $V \mapsto \mathcal{O}_X(\varphi^{-1}(V))$ for $V \subseteq Y$.

A locally ringed space is a ringed space where the stalk is a local ring.

A morphism of locally ringed spaces is a morphism of ringed spaces such that the induced ring homomorphism on the stalks is a local homomorphism, i.e. $\varphi^{-1}(\mathfrak{n}) = \mathfrak{m}$ (note that in general $\varphi^{-1}(\mathfrak{n}) \subseteq \mathfrak{m}$ always holds).

Theorem.

Suppose that $(X, \mathcal{O}_X)$ is a locally ringed space and A is a ring. Then there exists a (natural) bijection

$$\text{Hom}(X, \text{Spec}(A)) \cong \text{Hom}(A, \mathcal{O}_X(X))$$

given by

$$(\varphi, \varphi^\#) \longmapsto \varphi^\#(\text{Spec}(A))$$

where $\varphi^\# : \mathcal{O}_{\text{Spec}(A)}(\text{Spec}(A)) \rightarrow \varphi_* \mathcal{O}_X(\text{Spec}(A))$ is the induced pullback ring homomorphism, which is a ring homomorphism between $\mathcal{O}_{\text{Spec}(A)}(\text{Spec}(A)) = A$ and $\varphi_* \mathcal{O}_X(\text{Spec}(A)) = \mathcal{O}_X(X)$.

As a corollary, there exists a fully faithful contravariant functor from the category of affine schemes to the category of rings.

Surjectivity.

Suppose that $\alpha : A \rightarrow \mathcal{O}_X(X)$.

We construct a $\varphi : X \rightarrow \text{Spec}(A)$ such that $\alpha = \varphi^\#$.

$$\begin{array}{ccc}
 A & \xrightarrow{\alpha} & \mathcal{O}_X(X) \\
 & \searrow \text{re}_x \circ \alpha & \downarrow \text{re}_x \\
 & & \mathcal{O}_{X,x}
 \end{array}$$

Define $\varphi(x) := (\text{re}_x \circ \alpha)^{-1}(\mathfrak{m}_x) \in \text{Spec}(A)$.

We show that it is continuous:

To that end, it is sufficient to show that $\varphi^{-1}(U_f)$ is open in X for all $f \in A$, where $U_f := \{\mathfrak{p} \in \text{Spec}(A) \mid f \notin \mathfrak{p}\}$. Hence

$$\begin{aligned}
 \varphi^{-1}(U_f) &= \{x \in X \mid \varphi(x) \in U_f\} \\
 &= \{x \in X \mid f \notin (\text{re}_x \circ \alpha)^{-1}(\mathfrak{m}_x)\} \\
 &= \{x \in X \mid (\text{re}_x \circ \alpha)^{-1}(f) \notin \mathfrak{m}_x\} \\
 &= \{x \in X \mid (\text{re}_x \circ \alpha)(f) \in \mathcal{O}_{X,x}^\times\} \\
 &= \{x \in X \mid \exists h_x \in \mathcal{O}_{X,x} \text{ such that } h_x \cdot (\text{re}_x \circ \alpha)(f) = 1\} \\
 &= \{x \in X \mid \exists V_x \text{ open, } \exists h_x \in \mathcal{O}_X(V_x) \text{ such that } h_x \cdot [\alpha(f)]_x = 1\}
 \end{aligned}$$

such that $V_x \subseteq \varphi^{-1}(U_f)$, and it is open in X .

We now define what $\varphi^\# : \mathcal{O}_{\text{Spec}(A)} \rightarrow \varphi_* \mathcal{O}_X$ looks like on sheaves.

To that end, we first consider what $\varphi^\#$ might look like on the basic open sets U_f .

Specifically, we are looking to define a ring homomorphism between $\mathcal{O}_{\text{Spec}(A)}(U_f) = A[f^{-1}]$ and $(\varphi_* \mathcal{O}_X)(U_f) = \mathcal{O}_X(\varphi^{-1}(U_f))$ such that the following diagram commutes:

$$\begin{array}{ccc}
 A & \xrightarrow{\alpha} & \mathcal{O}_X(X) \\
 \downarrow & & \downarrow \text{res}_{\varphi^{-1}(U_f)} \\
 A[f^{-1}] & \xrightarrow{\alpha_{f^{-1}}} & \mathcal{O}_X(\varphi^{-1}(U_f))
 \end{array}$$

By the proof above, we can choose an open covering of U_f with open sets V_i such that $\alpha(f)$ is locally invertible on V_i . Let h_i denote this local inverse.

But then as inverses are unique, we get that $h_i|_{V_i \cap V_j} = h_j|_{V_i \cap V_j}$ for all i, j .

As $\mathcal{O}_X$ is a sheaf this patches uniquely to a global inverse $h \in \mathcal{O}_X(\varphi^{-1}(U_f))$.

So $[\alpha(f)]_{\varphi^{-1}(U_f)}$ is invertible in $\mathcal{O}_X(\varphi^{-1}(U_f))$, and by the universal property of localizations there exists a unique lift $\alpha_{f^{-1}}$ as desired.

We now consider what $\varphi^\#$ might look like on general open sets.

To that end, we show that this map is compatible with further restriction maps on basic open sets.

So suppose that $U_g \subseteq U_f$.

Note that this is true iff $g \in \sqrt{(f)}$, i.e. there exists a n and $a \in A$ such that $g^n = af$.

Hence there is an induced (A -linear) ring homomorphism $A[f^{-1}] \rightarrow A[g^{-1}]$ given by $\frac{1}{f} \mapsto \frac{a}{g^n}$.

$$\begin{array}{ccc}
 A & \xrightarrow{\alpha} & \mathcal{O}_X(X) \\
 \downarrow & & \downarrow \text{res}_{\varphi^{-1}(U_f)} \\
 A[f^{-1}] & \xrightarrow{\alpha_{f^{-1}}} & \mathcal{O}_X(\varphi^{-1}(U_f)) \\
 \downarrow & \searrow & \downarrow \text{res}_{\varphi^{-1}(U_g)} \\
 A[g^{-1}] & \xrightarrow{\alpha_{g^{-1}}} & \mathcal{O}_X(\varphi^{-1}(U_g))
 \end{array}$$

Notice that f is invertible under the composite map $\text{res}_{\varphi^{-1}(U_g)} \circ \alpha : A \rightarrow \mathcal{O}_X(\varphi^{-1}(U_g))$ such that by the universal property of localizations, the composite map uniquely factors through $A[f^{-1}]$ into the diagonal morphism.

But then $\alpha_{f^{-1}}$ and $\alpha_{g^{-1}}$ are morphisms such that they induce lifts of $\text{res}_{\varphi^{-1}(U_g)} \circ \alpha$ through $A[f^{-1}]$ as well; such that they satisfy the same universal property as the diagonal map.

Hence the diagram commutes. So φ is a morphism of sheaves and ringed spaces.

This result also extends $\varphi^\#$ to be a well-defined ring homomorphism on general open sets, as $\mathcal{O}_{\text{Spec}(A)}(U) = \varprojlim_{U_f \subseteq U} \mathcal{O}_{\text{Spec}(A)}(U_f)$ and $\varphi_* \mathcal{O}_X(U) = \varprojlim_{U_f \subseteq U} \varphi_* \mathcal{O}_X(U_f) = \varprojlim_{U_f \subseteq U} \mathcal{O}_X(\varphi^{-1}(U_f))$.

To complete the proof, we now show that $\varphi^\#$ is in fact a local morphism at all the stalks. Suppose that $\varphi(x) = \mathfrak{p}$, and consider the following commuting diagram.

$$\begin{array}{ccccc}
 A & \xrightarrow{\alpha} & \mathcal{O}_X(X) & & \\
 \downarrow & & \downarrow & \searrow \text{re}_x & \\
 A_{\mathfrak{p}} = \varprojlim_{f \notin \mathfrak{p}} A[f^{-1}] & \longrightarrow & \varprojlim_{U_f \ni \mathfrak{p}} \mathcal{O}_X(\varphi^{-1}(U_f)) & = & \mathcal{O}_{X,x}
 \end{array}$$

But then $\mathfrak{p} = (\text{re}_x \circ \alpha)^{-1}(\mathfrak{m}_x)$, so we have that $(\varphi^\#)^{-1}(\mathfrak{m}_x) = \mathfrak{p}$ in A .

Hence by the commutativity of the diagram we get that the preimage of $\mathfrak{m}_x$ in $A_{\mathfrak{p}}$ is exactly $\mathfrak{p}A_{\mathfrak{p}}$.

Injectivity.

Suppose that $\varphi, \psi : X \rightarrow \text{Spec}(A)$ such that $\varphi^\# = \psi^\#$.

Then for some fixed $x \in X$, let $\mathfrak{p} := \varphi(x)$ and $\mathfrak{q} := \psi(x)$ for some $\mathfrak{p}, \mathfrak{q} \in \text{Spec}(A)$.

Then by considering the commutative diagram

$$\begin{array}{ccc}
 A & \xrightarrow[\psi^\#]{\varphi^\#} & \mathcal{O}_X(X) \\
 \swarrow & \downarrow & \downarrow \text{re}_x \\
 A_{\mathfrak{p}} & \xrightarrow[\varphi_{\mathfrak{p}}^\#]{\psi_{\mathfrak{q}}^\#} & \mathcal{O}_{X,x}
 \end{array}$$

we get that $\mathfrak{p} = \mathfrak{q}$. So $\varphi = \psi$.